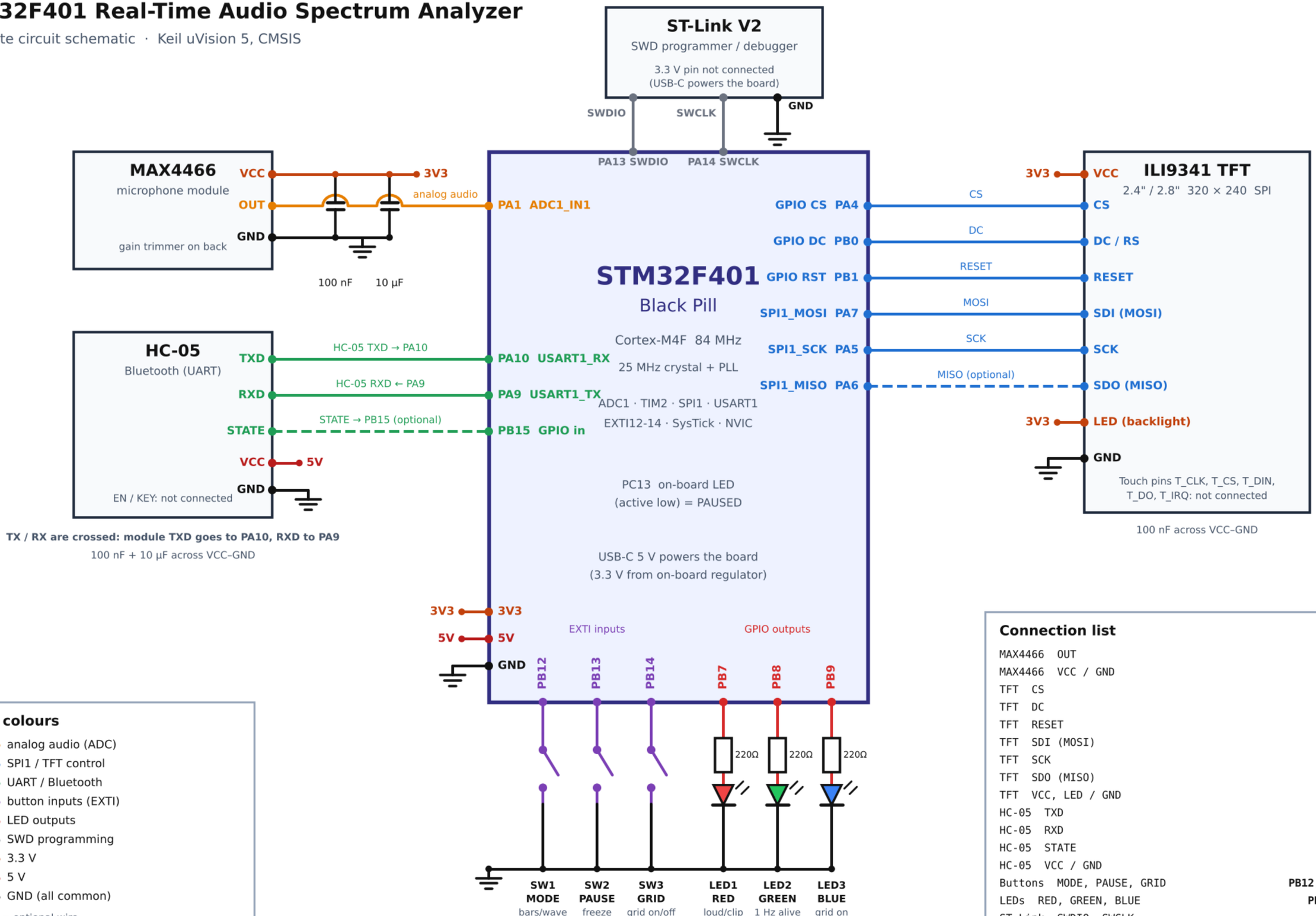
STM32F401 Real-Time Audio Spectrum Analyzer

Complete circuit schematic · Keil uVision 5, CMSIS



TX / RX are crossed: module TXD goes to PA10, RXD to PA9

100 nF + 10 µF across VCC–GND

Buttons: internal pull-up, press = GND (no resistor)

LED: long leg (anode) to the 220 Ω resistor, short leg (cathode) to GND

Wire colours

- analog audio (ADC)
- SPI1 / TFT control
- UART / Bluetooth
- button inputs (EXTI)
- LED outputs
- SWD programming
- 3.3 V
- 5 V
- GND (all common)

dashed = optional wire

Connection list

MAX4466 OUT	**PA1**
MAX4466 VCC / GND	**3V3 / GND**
TFT CS	**PA4**
TFT DC	**PB0**
TFT RESET	**PB1**
TFT SDI (MOSI)	**PA7**
TFT SCK	**PA5**
TFT SDO (MISO)	**PA6 (opt.)**
TFT VCC, LED / GND	**3V3 / GND**
HC-05 TXD	**PA10**
HC-05 RXD	**PA9**
HC-05 STATE	**PB15 (opt.)**
HC-05 VCC / GND	**5V / GND**
Buttons MODE, PAUSE, GRID	**PB12, PB13, PB14**
LEDs RED, GREEN, BLUE	**PB7, PB8, PB9**
ST-Link SWDIO, SWCLK	**PA13, PA14**

System Block Diagram and Signal Chain

From sound to bars: every stage, with the numbers to remember

TIM2 timer trigger
16 kHz, one trigger every 62.5 µs

starts each conversion

1 **Sound**
voice, music, test tone

2 **MAX4466 microphone**
electret + amplifier
output centred at 1.65 V

3 **ADC1 (pin PA1)**
12-bit: 0 to 4095
silence is about 2048

4 **ADC interrupt**
ADC_IRQHandler, NVIC priority 1
stores 1 sample per trigger

5 **Sample buffer**
512 samples = 32 ms
then flag s_cap_ready = 1

Sampling rate Fs = 16 kHz → bandwidth = Fs / 2 = 8 kHz (Nyquist)

main loop: spectrum_load() → spectrum_compute()

6 **DC removal**
subtract the mean
clip check + RMS level

7 **Hann window**
fades the block edges
less spectral leakage

8 **512-point FFT**
radix-2, 9 stages
31.25 Hz per bin

9 **Power → 32 bars**
log-spaced, 31 Hz to 8 kHz
converted to dB (dBFS)

10 **Peak and level**
peak frequency (interpolated)
loudness in %, CLIP flag

ui_update()

bt_report() every 250 ms

11 **ILI9341 TFT**
SPI1 at 10.5 MHz
bars, peak dots, waveform, text

12 **Phone over Bluetooth**
USART1 → HC-05, 9600 baud
F=...Hz L=...% B=32 digits

13 **Status LEDs**
red PEAK · green HEARTBEAT
blue GRID · on-board PAUSED

LEDs

14 **MAIN LOOP (main.c)**
reads buttons and phone commands → mode, pause, grid, gain
decides what to draw, send and light up

15 **3 push buttons**
PB12 MODE · PB13 PAUSE · PB14 GRID
EXTI interrupt, NVIC priority 4

16 **Phone commands**
M P G + - ?
USART1 interrupt, NVIC priority 3

Numbers to remember

CPU 84 MHz · Fs 16 kHz · bandwidth 8 kHz
FFT 512 points · 31.25 Hz per bin · block 32 ms
32 log bars: 31 Hz to 8 kHz · plot range −75 to −15 dB
TFT SPI 10.5 MHz · Bluetooth 9600 baud
NVIC priorities: ADC 1 · SysTick 2 · USART1 3 · EXTI 4

Firmware Flowchart: Start-up, Main Loop and Interrupts

Interrupt handlers only store data or set flags. The heavy work (FFT, drawing) runs in the main loop.

START-UP (runs once)

1. **Enable the FPU**
   CPACR bits for CP10 / CP11
2. **bsp_clock_init()**
   84 MHz from 25 MHz crystal (or HSI)
3. **bsp_systick_init()**
   1 ms tick
4. **bsp_io_init()**
   LEDs outputs, HC-05 STATE input
5. **bt_init(9600)**
   USART1 + receive interrupt
6. **lcd_init()**
   SPI1 + ILI9341 start-up table
7. **spectrum_init()**
   window, FFT and bar tables
8. **ui_splash(), wait 1.5 s**
   splash screen, message to phone
9. **ui_init()**
   draw the main screen
10. **bsp_buttons_init()**
    EXTI12-14 + NVIC
11. **bsp_sampling_init() / start**
    TIM2 + ADC1 + NVIC, first capture

MAIN LOOP (repeats forever)

a. **bsp_buttons_poll(now)**
   confirm a press 25 ms later, ignore bounce for 150 ms

b. **bt_getc() → bt_command()**
   phone letters M P G + - ? become the same events

c. **Handle events**
   MODE: bars ↔ wave PAUSE: freeze + PC13 LED
   GRID: grid on/off + blue LED message to phone

512-sample block ready?
- no → skip to f
- yes →

d. **spectrum_load() + spectrum_compute()**
   copy block, remove DC, THEN bsp_sampling_start()
   (next block records while the FFT runs)
   window → FFT → 32 bars + peak + level

paused?
- yes → skip to f
- no →

e. **Draw and send**
   ui_update(): bars / waveform on the TFT (SPI1)
   ui_peak_alert() → red LED for 150 ms
   bt_report() every 250 ms → phone

f. **ui_refresh() and LEDs**
   redraw text if changed, update PEAK LED
   toggle HEARTBEAT LED every 500 ms

repeat forever

INTERRUPTS (run in the background)

EXTI15_10_IRQHandler
NVIC priority 4
falling edge on PB12-14: record press time, set s_btn_pend

s_btn_pend → bsp_buttons_poll(now)

USART1_IRQHandler
NVIC priority 3
RX byte → receive ring buffer
TX byte ← transmit ring buffer

rx ring buffer ↔ bt_getc() / bt_command()

ADC_IRQHandler
NVIC priority 1 (most urgent)
TIM2 triggers ADC1 every 62.5 µs
s_cap[i++] = ADC1->DR
after 512 samples: stop TIM2, s_cap_ready = 1

s_cap_ready → 512-sample block ready?

SysTick_Handler
NVIC priority 2
every 1 ms: s_ms++ → millis(), delay_ms()

millis() → ui_refresh() and LEDs

HardFault_Handler
CPU crash: blink the on-board LED fast forever (so a crash is visible)

How to read this chart

- program flow (main loop)
- flag or buffer shared with an interrupt (volatile)

Rule: an interrupt handler stays short; the FFT and drawing are done in the main loop.